$$E(X) = p + \frac{p}{3} + p^2 = \frac{4}{3}$$

$$\implies (3p - 2)(p + 2) = 0$$

$$\because 1 > p > 0$$
$$\therefore p = \frac{2}{3}$$

Guilty-verdict probabilities: $\frac{2}{3}, \frac{2}{9}, \frac{4}{9}$
Let G be the probability that at least two judges give guilty verdicts.

$$\begin{aligned} G &= \frac{2}{3} \cdot \frac{2}{9} \cdot \frac{5}{9} + \frac{2}{3} \cdot \frac{4}{9} \cdot \frac{7}{9} + \frac{2}{9} \cdot \frac{4}{9} \cdot \frac{1}{3} + \frac{2}{3} \cdot \frac{2}{9} \cdot \frac{4}{9} \\ &= \frac{20}{243} + \frac{56}{243} + \frac{8}{243} + \frac{16}{243} \\ &= \frac{100}{243} \end{aligned}$$

Defendant guilty with probability t.

$$\begin{aligned} P(correct) &= t \cdot G + (1 - t) \cdot (1 - G) \\ &= \frac{143 - 43t}{243} \end{aligned}$$

$$\because P(correct) = \frac{1}{2}$$

$$286 - 86t = 243 \implies t = \frac{1}{2}$$

This solution was submitted by **Jon**
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